

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is: ***An issue with the TCP handshake protocol causing a communication issue with the webserver.***

The logs show that: ***A suspicious source IP 203.0.113.0 is sending multiple SYN requests to the destination IP causing the travel agency's website to become overloaded with requests and unable to function.***

This event could be: ***Dos (Denial-of-Service) attack via SYN flooding.***

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

- 1. The source IP will send over a SYN or synchronize request establish a connection to the web server***
- 2. Once the SYN request has been received, the web server would respond back with a SYN ACK or synchronized-acknowledgement letting the source IP know that it acknowledges the request to establish a connection.***
- 3. To finalize the connection, the source will reply back to the web server ACK or acknowledge to begin streaming data.***

Explain what happens when a malicious actor sends a large number of SYN packets all at once: ***The large number of SYN packets sent in rapid sequence will cause the web server to be overwhelmed and exhaust the connection backlog queue resulting in customers and employees of the travel agency receiving an error page when trying to connect to the website.***

